

```
In [1]: import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures, StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.metrics import mean_squared_error, r2_score, mean_absolute_error
import matplotlib.pyplot as plt
```

```
In [2]: df=pd.read_csv('encoded_car_data (1).csv')
df.head()
```

Out[2]:

	diesel	gas	std	turbo	convertible	hardtop	hatchback	sedan	wagon	4wd	...	wheelbase
0	0.0	1.0	1.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	...	88.6
1	0.0	1.0	1.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	...	88.6
2	0.0	1.0	1.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0	...	94.5
3	0.0	1.0	1.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0	...	99.8
4	0.0	1.0	1.0	0.0	0.0	0.0	0.0	1.0	0.0	1.0	...	99.4

5 rows × 36 columns



```
In [4]: x=df[['engine size', 'horsepower', 'citympg', 'highwaympg']]
y=df['price']
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=
lr=Pipeline([
    ('scaler', StandardScaler()),
    ('model', LinearRegression())
])
```

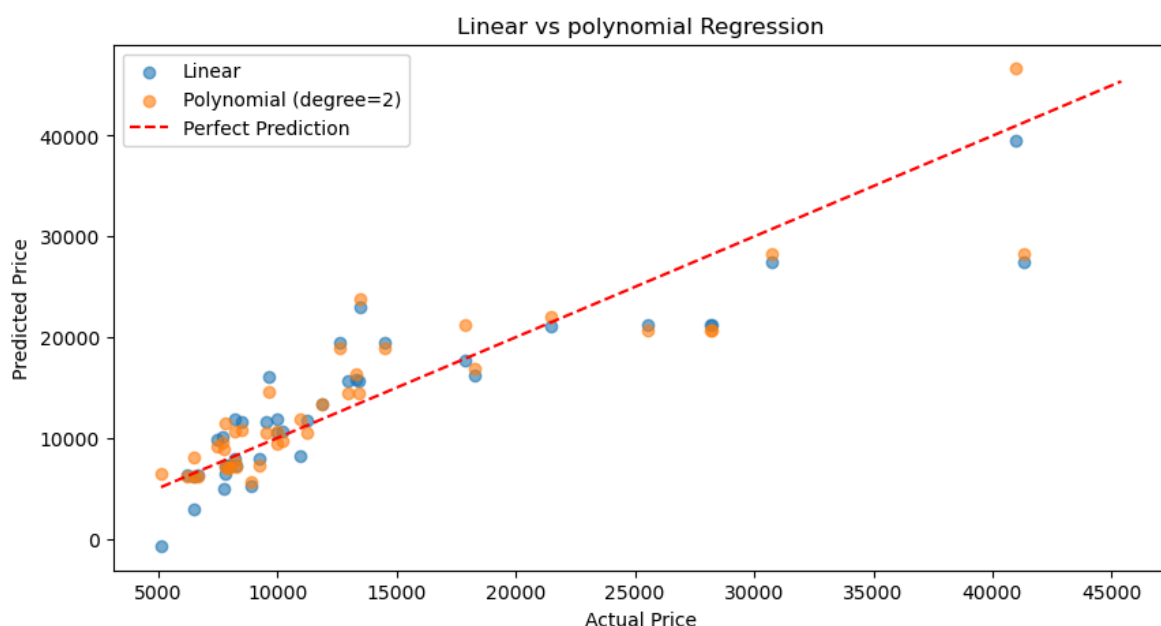
```
In [5]: lr.fit(x_train, y_train)
y_pred_linear=lr.predict(x_test)
poly_model=Pipeline([
    ('poly', PolynomialFeatures(degree=2)),
    ('scaler', StandardScaler()),
    ('model', LinearRegression())
])
```

```
In [6]: poly_model.fit(x_train, y_train)
y_pred_poly=poly_model.predict(x_test)
print('Name: RAGHUL.S')
print('Reg.No: 212225040325')
print("Linear Regression")
print('MSE=',mean_squared_error(y_test,y_pred_linear))
print('MAE=',mean_absolute_error(y_test,y_pred_linear))
r2score=r2_score(y_test,y_pred_linear)
print('R2 Score=',r2score)
```

Name: RAGHUL.S
 Reg.No: 212225040325
 Linear Regression
 MSE= 16471505.900042146
 MAE= 2892.628134137953
 R2 Score= 0.7913520781370976

```
In [7]: print("\nPolynomial Regression:")
print(f"MSE: {mean_squared_error(y_test,y_pred_poly):.2f}")
print(f"MAE: {mean_absolute_error(y_test,y_pred_poly):.2f}")
print(f"R²: {r2_score(y_test, y_pred_poly):.2f}")
plt.figure(figsize=(10,5))
plt.scatter(y_test, y_pred_linear, label='Linear', alpha=0.6)
plt.scatter(y_test,y_pred_poly, label='Polynomial (degree=2)',alpha=0.6)
plt.plot([y.min(),y.max()], [y.min(),y.max()], 'r--',label='Perfect Prediction')
plt.xlabel("Actual Price")
plt.ylabel("Predicted Price")
plt.title("Linear vs polynomial Regression")
plt.legend()
plt.show()
```

Polynomial Regression:
 MSE: 15247661.89
 MAE: 2694.05
 R²: 0.81



In []: